

MEASURING AND ENHANCING TRUSTWORTHINESS OF LLMS IN RAG THROUGH GROUNDED ATTRIBUTIONS AND LEARNING TO REFUSE

Maojia Song, Shang Hong Sim, Rishabh Bhardwaj, Hai Leong Chieu, Navonil Majumder, Soujanya Poria

Singapore University of Technology and Design
DSO National Laboratories

May 24, 2026

Navigation icons

Maojia Song et al. (SUTD)

Measuring and Enhancing Trustworthiness...

May 24, 2026

1 / 21

Background: Why RAG Needs Grounding

- LLMs excel at language but struggle with factual reliability in knowledge-heavy tasks.
- RAG grounds answers in external documents, boosting factuality and verifiability.
- Attributable RAG requires inline citations so users can check claims.

Navigation icons

Maojia Song et al. (SUTD)

Measuring and Enhancing Trustworthiness...

May 24, 2026

2 / 21

Problem: Trustworthy LLMs in RAG

- End-to-end RAG evaluations conflate retriever strength with LLM grounding ability.
- We need LLMs that: answer only from given docs, refuse when evidence is missing, and cite correctly.
- Prompting trades off over- vs under-responsiveness; weak on true grounding.

Navigation icons

Maojia Song et al. (SUTD)

Measuring and Enhancing Trustworthiness...

May 24, 2026

3 / 21

Contributions Overview

- TRUST-SCORE: holistic, LLM-centric trustworthiness metric for RAG (refusals, answers, citations).
- TRUST-ALIGN: data-centric alignment pipeline targeting RAG-specific hallucination modes.
- Direct Preference Optimization (DPO) over 19K preference pairs yields broad gains across models/datasets.

Navigation icons

Maojia Song et al. (SUTD)

Measuring and Enhancing Trustworthiness...

May 24, 2026

4 / 21

- Three components: F1GR (grounded refusals), F1AC (calibrated answer correctness), F1GC (grounded citations).
- Entailment and answerability checked via TRUE-based NLI to avoid substring bias.
- Consolidated score avoids parametric-knowledge gaming and retriever confounds.

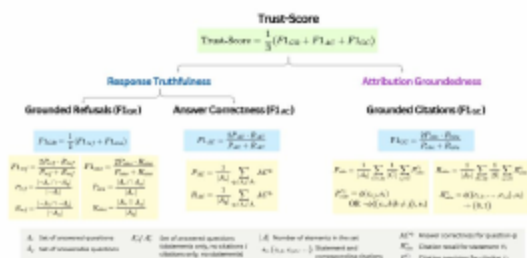


Figure: TRUST-SCORE calculation shown as a computational graph.

F1GR: Grounded Refusals

- Reward correct refusal when docs don't entail any gold claim.
- Reward correct answers when docs support them.
- Balances over-responsiveness (answering without evidence) vs excessive refusal.

F1AC: Calibrated Answers

- Compare responses to the subset of gold claims derivable from the provided docs.
- Precision- and recall-oriented aggregations reduce gaming by verbosity or guesswork.
- TRUE-NLI guards against superficial string overlap.

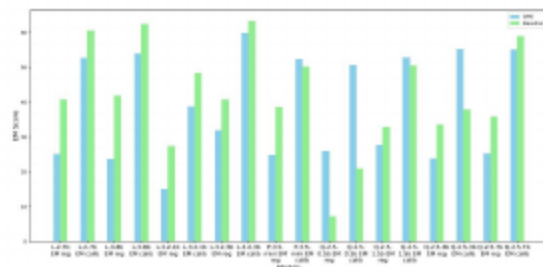


Figure: Calibrated vs regular answer correctness (ASQA).

F1GC: Grounded Citations

- Citation recall: do cited docs collectively support the statement?
- Citation precision: are cited docs necessary/supportive (avoid over-citation)?
- Combine into F1 to capture faithfulness and sufficiency of attributions.

- Naive substring matching mistakes topical overlap for entailment.
- TRUE-based NLI marks questions unanswerable when docs don't entail gold claims.
- Reduces false positives in answerability, enabling fair F1GR/F1AC computation.
- Substring overlap mislabels answerability; TRUE-NLI correctly flags unanswerable contexts, reducing false positives.

- Build preference data targeting five RAG-specific hallucination modes.
- Generate grounded preferred responses and mine hard negative responses.
- Optimize models with DPO to align toward grounded refusal, answers, and citations.



Figure: TRUST-ALIGN pipeline overview.

- Retrieve and select oracle docs; recombine relevant/irrelevant passages.
- Induce answerable and unanswerable contexts for the same question.
- Shuffle doc order to reduce index bias in citations.

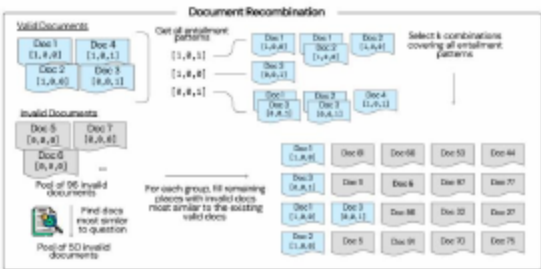


Figure: Document recombination process in augmented prompt curation.

- Synthesized r+ by stitching calibrated gold claims into natural responses.
- Map each statement to a minimal sufficient subset of supporting documents.
- Explicitly assign refusals when contexts are unanswerable.



Figure: Claim-document-mapping process.

- Run an SFT model to generate candidate r ; detect five error types.
- Score severity and select top 50% hardest negatives.
- Targeted coverage: inaccurate answers, over-responsiveness, excessive refusals, over-/improper citation.

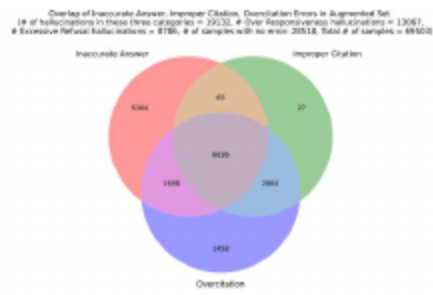


Figure: Hallucination statistics for hard-negative mining.

- We quantify error frequencies and severities to prioritize difficult negatives for alignment.
- Most frequent errors: inaccurate claims, over-responsiveness, over-citation; these informed hard-negative mining.

- Model-agnostic alignment using paired $(q, D, r+, r)$.
- DPO surpasses SFT in 26/27 settings, especially citations (F1GC).
- Teaches refusal correctness without over-reliance on parametric knowledge.

- Datasets: ASQA, QAMPARI, ELI5; OOD: ExpertQA (answerable/unanswerable mixes).
- Metrics: TRUST-SCORE (F1GR, F1AC, F1GC), AR% (responsiveness).
- Models: LLaMA (1–8B), Qwen2.5 (0.5–7B), Phi-3.5.
- Baselines: ICL (Brown et al., 2020); Self-RAG (Asai et al., 2023); FRONT (Zhou et al., 2024); PostCite (ours); PostAttr (ours).

Main Results

- 26/27 configs: higher TRUST-SCORE than baselines.
- LLaMA-3-8B vs FRONT: +12.56 (ASQA), +36.04 (QAMPARI), +17.69 (ELI5).
- Consistent gains in F1GR (refusals) and F1GC (citations); strong F1AC on QAMPARI.

Navigation icons

Generalization: ExpertQA

- TRUST-ALIGNeD LLaMA-3-8B achieves TRUST 54.85, surpassing GPT-4 ICL (54.69).
- Outperforms FRONT across open-weight families on ExpertQA.
- Shows robustness to domain shifts with better refusal and citation grounding.

Navigation icons

Closed-Source Comparison

- TRUST-ALIGNeD open models rival or surpass closed ICL on TRUST-SCORE.
- Better refusal correctness and grounded citations with lower parametric reliance.
- Demonstrates practical viability of small open-weight models for trustworthy RAG.

Table: Table 12: Our models vs closed source: AR% := Answered Ratio in %; F1AC := Answer Correctness F1 (Calibrated); F1GR := Grounded refusals F1; F1GC := Grounded Citation F1; TRUST := TRUST score. R := Refusal prompt is used. D := Default prompt is used.

Model	Prompt	AR%	F1AC	F1GR / F1GC	TRUST
ICL-GPT-3.5	R	71.20	–	52.91 / 83.94	67.64
ICL-GPT-4	R	86.81	–	62.96 / 84.35	69.72
ICL-Claude-3.5	R	84.60	–	59.97 / 68.35	64.36
ICL-GPT-3.5	D	94.41	–	55.03 / 78.04	61.85
ICL-GPT-4	D	92.72	–	62.37 / 79.70	65.41
ICL-Claude-3.5	D	82.49	–	54.20 / 58.88	59.86

Limitations

- Dependence on NLI quality for answerability/entailment judgments.
- Granularity of citation scoring may miss intra-document span-level errors.
- Prompt-only methods unstable; alignment requires curated data and compute.

Navigation icons

- How to make claim decomposition more robust for long-form answers?
- Can we calibrate responsiveness (reduce RAC bias) without hurting recall?
- How sensitive is TRUST-SCORE to entailment quality, and can stronger multi-hop NLI help?
- How to co-train retriever-generator while preserving LLM-centric evaluation?
- Can span-level citation scoring and scalable hard-negative selection improve groundedness further?